OPT as a Cognitive Routing Layer for Modular LLM Architectures

Abstract

Recent large language models increasingly rely on modular architectures to achieve scalability and efficiency. Among these approaches, Mixture-of-Experts (MoE) systems enable conditional computation by dynamically routing inputs to specialized expert modules. Although this mechanism allows models to expand parameter capacity while maintaining manageable computational cost, the routing processes that determine expert activation remain largely opaque. Current gating mechanisms primarily optimize statistical performance, offering limited insight into the structural logic governing how information flows through the system.

This paper proposes a conceptual framework in which the Ontology–Process–Trajectory (OPT) model of cognition functions as a cognitive routing layer for modular large language model architectures. The OPT framework describes reasoning processes as dynamic pathways connecting generative signal sources with stabilization processes that resolve those signals into coherent interpretations or actions. By introducing this pathway perspective into modular AI architectures, expert routing can be interpreted as structured signal propagation rather than purely statistical gating.

Within this perspective, modular experts correspond to stabilization mechanisms that transform signals within particular domains of processing, while routing mechanisms approximate the selection of cognitive pathways through which signals travel. The resulting architecture suggests a layered interpretation in which neural computation operates within a structural routing topology that organizes information flow.

The present work does not introduce a new training algorithm. Instead, it proposes a structural interpretation of modular neural architectures and outlines how pathway-based routing principles could inform future model design. By linking cognitive pathway theory with modular machine learning systems, the OPT framework may offer a conceptual bridge between theories of reasoning and scalable artificial intelligence architectures.

1 Introduction

The rapid development of large language models has led to a growing interest in modular neural architectures that allow systems to scale beyond the limits of fully dense networks (Naveed et al., 2025). As model sizes increase into the hundreds of billions or trillions of parameters, conditional computation mechanisms have become essential for maintaining computational efficiency. Instead of activating all parameters for every input, modular architectures selectively route information to specialized computational units that perform specific transformations (Lepikhin et al., 2020).

Among these approaches, Mixture-of-Experts (MoE) architectures have emerged as one of the most influential solutions (Li et al., 2025; Shazeer et al., 2017). In MoE systems, a routing network dynamically selects a subset of expert modules for each token or representation. This mechanism allows models to increase parameter capacity while keeping the computational cost per inference step relatively stable.

Despite their practical success, MoE architectures introduce an interpretability challenge (Dimitri, Regina, & Alfonz, 2025). The routing networks that determine expert activation are typically trained through gradient-based optimization without explicit structural constraints. As a result, the internal logic governing how information is routed through the system often remains opaque. While the architecture itself is modular, the reasoning processes that determine which modules are activated are difficult to interpret.

This challenge reflects a broader limitation in current machine learning systems (Ling et al., 2025). Neural networks often perform complex reasoning tasks while providing limited insight into how information flows through their internal components (Samek, Montavon, Lapuschkin, Anders, & Müller, 2021). Although interpretability research has developed techniques for analysing attention mechanisms and representation spaces, the structural interpretation of routing mechanisms remains comparatively underdeveloped.

At the same time, cognitive science has long emphasized the importance of structured pathways in reasoning processes (Beer, 2000). Human cognition rarely proceeds through uniform computation across all available processes. Instead, signals are selectively propagated through pathways that connect generative processes with stabilization mechanisms that interpret and integrate those signals. From this perspective, reasoning emerges not simply from representations but from the routes through which information travels.

The Ontology–Process–Trajectory (OPT) framework was developed to describe cognition in precisely these structural terms (Liu, 2026). Instead of categorizing individuals into fixed psychological types, the framework models cognition as the movement of signals through pathways that connect generative signal sources with stabilization sinks. Within this model, reasoning unfolds through dynamic routing processes that reorganize pathways in response to internal constraints and environmental inputs.

This paper explores how such a pathway perspective can be applied to modular neural architectures. Specifically, we propose that the OPT framework can be interpreted as a cognitive routing layer that provides structural meaning to the routing mechanisms already present in architectures such as Mixture-of-Experts models. Rather than replacing existing neural routing systems, the framework offers a conceptual structure that interprets expert activation as the propagation of signals through cognitive pathways.

Under this interpretation, modular experts correspond to stabilization processes that resolve signals within particular computational domains, while routing networks approximate the selection of pathways that connect generative representations to these stabilization processes. The resulting perspective suggests that large language models may already operate within latent pathway topologies that resemble structured reasoning systems.

The goal of the present work is therefore not to propose a new training algorithm but to articulate a structural framework that links modular neural architectures with cognitive pathway theory. By introducing the concept of a cognitive routing layer, the OPT framework may provide a conceptual bridge between theories of reasoning and scalable artificial intelligence systems.

2 Modular Architectures in Large Language Models

The scaling of modern language models has required architectural innovations that allow neural networks to increase parameter capacity without proportionally increasing computational cost. One of the central strategies used to achieve this goal is modular computation (Auda & Kamel, 1999).

In modular neural architectures, different parts of the model are responsible for processing different kinds of signals or representations (Yao, Xu, Zhang, Liang, & Li, 2020). Instead of performing identical transformations across all components of the network, modular systems allow specialized units to handle particular patterns of information. This specialization enables more efficient use of parameters and may allow models to develop distinct competencies across different domains of reasoning.

Mixture-of-Experts architectures represent one of the most widely used implementations of this modular approach (Fedus, Zoph, & Shazeer, 2022). In an MoE layer, multiple expert networks operate in parallel, while a routing mechanism determines which experts should process a given input. For an input representation x, the output of the layer can be written as

$$y = \sum_{i=1}^{n} g_i\,(x) E_i(x)$$

where $E_i$represents the transformation performed by the i-th expert and $g_i(x)$represents the routing weight assigned by the gating network.

In practice, many implementations use sparse routing strategies in which only a small number of experts are activated for each token. This design allows models to maintain extremely large parameter counts while keeping computational requirements manageable during inference.

Although modular architectures enable efficient scaling, they also introduce new questions regarding the interpretation of internal model dynamics (Ling et al., 2025; Rosenbaum, Cases, Riemer, & Klinger, 2019). Expert modules may become specialized during training, but the structural logic that governs routing decisions often remains difficult to analyse. The gating network learns routing behaviour through gradient optimization, yet its decisions are not typically constrained by explicit structural principles.

As a result, understanding how information flows through modular neural systems remains an open problem. While the architecture itself provides discrete components, the reasoning processes that connect those components are not always transparent.

This observation motivates the search for structural frameworks that can help interpret routing behaviour in modular models. One possible approach is to view routing not simply as statistical selection but as the movement of signals through structured pathways within a computational system.

3 The OPT Cognitive Pathway Model

The Ontology–Process–Trajectory (OPT) framework proposes a structural model of cognition in which reasoning processes are understood as the propagation of signals through organized pathways rather than as the expression of static cognitive traits (Liu, 2026). Within this perspective, cognitive activity emerges from the interaction between signal-generating mechanisms and stabilization processes that resolve those signals into coherent interpretations, decisions, or actions.

The model conceptualizes cognition as a routing system consisting of two functional layers: generative sources and stabilization sinks. Signals originate from generative sources that produce candidate representations of meaning, and these signals propagate through pathways toward stabilization processes that integrate, validate, or operationalize them. A reasoning episode can therefore be interpreted as a trajectory connecting the origin of a signal with the

process that ultimately stabilizes its interpretation. The overall structure of the OPT framework is illustrated in Figure 1.

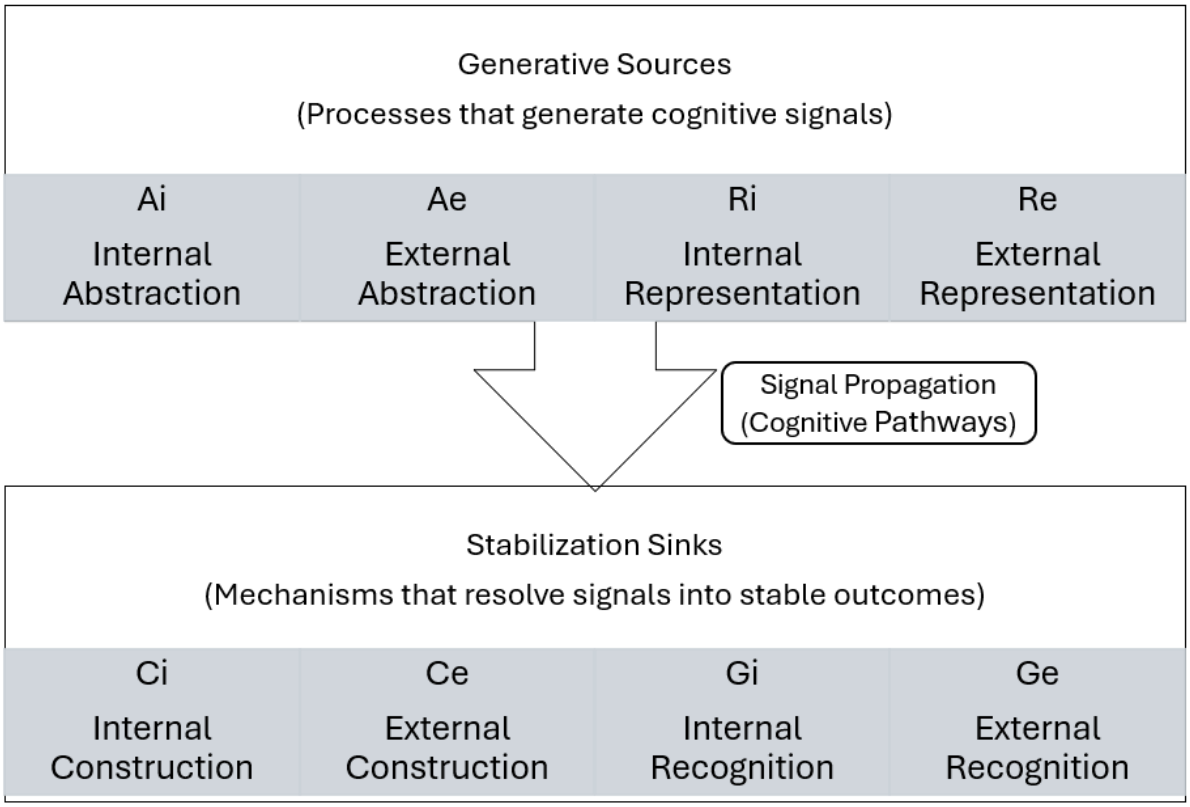

Figure 1. Architecture of the OPT cognitive pathway framework.

Generative sources produce cognitive signals through abstraction or representational retrieval processes. These signals propagate through cognitive pathways and are resolved by stabilization sinks that integrate them into structured interpretations, decisions, or actions.

Formally, the OPT framework represents a cognitive pathway as a mapping between a source unit and a sink unit:

$$p = (s_i, t_j)$$

where s_i denotes a generative source and t_j denotes a stabilization sink. Each reasoning event can therefore be interpreted as the activation of a pathway p that connects signal generation with signal stabilization.

The framework distinguishes four types of generative sources:

$$S = \{Ai, Ae, Ri, Re\}$$

where Ai represents internal abstraction processes that generate structural patterns and conceptual directions, Ae represents external abstraction processes that produce associative expansions in response to environmental stimuli, Ri denotes internal representation processes that retrieve experience-based knowledge from memory, and Re denotes external representation processes that track real-time environmental signals and situational feedback.

Correspondingly, the framework defines a set of stabilization sinks:

$$T = \{Ci, Ce, Gi, Ge\}$$

where Ci represents internal construction processes that integrate signals into internally coherent conceptual structures, Cedenotes external construction processes that translate signals into organized action, Girepresents internal recognition processes that stabilize signals through confirmation of internal coherence or personal meaning, and Gedenotes external recognition processes that stabilize signals through alignment with social or relational contexts.

A cognitive pathway can therefore be described as a trajectory through the space defined by the Cartesian product of source and sink sets:

$$P = S \times T$$

where each element $p \in P$represents a possible reasoning pathway connecting a generative process with a stabilization mechanism.

Importantly, cognitive activity does not involve the activation of a single fixed pathway. Instead, reasoning unfolds through the dynamic reconfiguration of pathway activation patterns. At a given moment, the cognitive system can therefore be described by a pathway activation distribution:

$$\pi(p \mid x)$$

which represents the probability that pathway $p$becomes active given an input signal x.

Within this framework, reasoning processes correspond to the propagation and stabilization of signals across the pathway space defined by P. Environmental input, internal constraints, and prior representations all influence the routing distribution that determines which pathways become active.

This formulation emphasizes that cognition involves not only representations but also routing dynamics that govern how signals move through the system. Rather than assuming that reasoning emerges from uniform processing across all components, the OPT framework models cognition as the structured movement of signals through a pathway topology.

Such a pathway perspective provides a structural account of how reasoning processes evolve over time and establishes a formal basis for interpreting routing behaviour in modular computational architectures.

4 OPT as a Cognitive Routing Layer for Modular LLM Architectures

The pathway formulation introduced in the previous section provides a structural basis for interpreting routing behaviour in modular neural architectures. Large language models increasingly rely on conditional computation mechanisms in which signals are selectively routed toward specialized processing modules (Wang, Li, Chen, Zhang, & Zhu, 2025). Within the OPT framework, such routing behaviour can be interpreted as the activation of pathways that connect signal-generating processes with stabilization mechanisms inside the computational system.

In modular architectures such as Mixture-of-Experts (MoE) models, a routing network determines which expert modules process a given input representation. For an input signal x,

the gating network produces a routing distribution across experts, and the output of the layer is computed as a weighted combination of expert responses:

$$y = \sum_{k=1}^{n} g_k\,(x) E_k(x)$$

where E_k denotes the transformation performed by the k-th expert and $g_k(x)$ represents the routing weight assigned by the gating network.

Within the OPT interpretation, this routing distribution can be viewed as an approximation of the pathway activation distribution introduced earlier:

$$g_k(x) \approx \pi(p \mid x)$$

In other words, the gating mechanism of a modular neural system can be interpreted as a computational approximation of pathway routing within a structured signal space. Instead of selecting experts purely through statistical optimization, the routing system implicitly determines which stabilization mechanisms should process the signals generated by the model.

Under this perspective, expert modules correspond to stabilization sinks that resolve incoming signals into transformed representations. Different experts may specialize in stabilizing different classes of signals, effectively acting as computational nodes within the pathway topology defined by the OPT framework.

The structural correspondence between modular neural architectures and the OPT pathway model can be summarized as follows.

Table 1 Structural Correspondence Between OPT Pathways and Modular LLM Architectures

Modular LLM Component	OPT Interpretation	Functional Role
Input token representation	Signal x	Initial signal entering the routing system
Internal representation layers	Generative sources S	Generation of candidate signal structures
Routing / gating network	Pathway distribution $\pi(p \mid x)$	Selection of signal propagation routes
Expert modules	Stabilization sinks T	Transformation and stabilization of signals
Model output	Stabilized representation	Final resolved signal after pathway processing

This mapping suggests that modular language models can be interpreted as systems in which signals propagate through a latent pathway topology. The routing network determines how signals travel through this topology, while expert modules perform the stabilization transformations that ultimately shape the model's output.

Such an interpretation does not imply that existing neural systems explicitly implement the OPT pathway model. Rather, it proposes that the statistical routing mechanisms learned during training may approximate structural routing dynamics similar to those described by the framework. Viewing modular architectures through this lens allows routing behaviour to be interpreted as structured signal propagation rather than purely statistical gating.

This perspective also introduces the concept of a cognitive routing layer. Instead of treating routing mechanisms as auxiliary components of neural architectures, the OPT framework interprets them as structural processes that organize how signals move through the computational system. The routing layer therefore determines not only which modules process a signal but also how reasoning trajectories unfold across the network.

Recognizing routing mechanisms as cognitive pathway selectors may provide new perspectives on how modular neural systems develop specialized reasoning capabilities. Expert specialization may emerge when certain pathways are repeatedly activated during training, allowing particular experts to become associated with specific stabilization roles within the system.

From this viewpoint, the modular structure of large language models resembles a distributed reasoning architecture in which signals move through multiple potential pathways before stabilizing as outputs. The OPT framework provides a structural vocabulary for describing these pathways and for analysing how routing dynamics shape the behaviour of large-scale neural systems.

5 Pathway-Constrained Routing Hypotheses

Interpreting modular neural architectures through the OPT pathway framework suggests that routing behaviour in large language models may exhibit structural regularities that resemble cognitive pathway dynamics. Although current architectures do not explicitly implement the pathway topology defined by the OPT model, the repeated routing of signals during training may give rise to emergent routing structures that approximate such pathways.

This section outlines several hypotheses that follow from interpreting modular language models as systems operating within a latent pathway space. These hypotheses do not assume that existing models explicitly encode the OPT architecture. Instead, they propose that observable routing patterns in modular neural systems may reflect structural tendencies that can be analysed using the pathway framework.

5.1 Pathway Specialization Hypothesis

The first hypothesis concerns the emergence of expert specialization. In Mixture-of-Experts architectures, expert modules are trained through repeated exposure to routed signals. Over time, this repeated routing may cause certain experts to become associated with particular classes of signal structures.

Within the OPT interpretation, this process can be viewed as the stabilization of specific pathways. Signals originating from similar generative conditions may repeatedly activate the same stabilization processes, causing particular experts to specialize in resolving certain signal types.

If this hypothesis holds, expert activation patterns should not be random across tasks. Instead, experts may display stable activation tendencies when processing particular reasoning contexts

or representational structures. In such cases, expert modules effectively function as stabilization sinks within the pathway topology.

Empirical investigation of this hypothesis could involve analysing expert activation frequencies across tasks and identifying whether consistent routing patterns emerge for different categories of reasoning.

5.2 Routing Stability Hypothesis

A second hypothesis concerns the stability of routing distributions during inference. If routing mechanisms approximate pathway selection processes, then repeated reasoning contexts may produce stable routing patterns across similar inputs.

In current implementations, gating networks determine routing distributions based on learned transformations of input representations. Under the pathway interpretation, these distributions can be understood as approximations of pathway activation probabilities.

If routing decisions reflect structural signal propagation rather than purely local statistical optimization, then similar reasoning contexts may activate comparable routing patterns across multiple inference steps. This would suggest that the routing system organizes signals through relatively stable pathway trajectories rather than through isolated expert selections.

Analysing routing stability across repeated reasoning tasks could therefore provide insight into whether modular neural systems exhibit pathway-like routing behaviour.

5.3 Interpretable Expert Topology Hypothesis

A third hypothesis concerns the possibility that modular architectures may develop interpretable expert topologies. If experts correspond to stabilization processes within a pathway system, then the distribution of expert specialization may form a structured topology across the model.

Instead of acting as independent modules, experts may occupy functional positions within a distributed stabilization network. Some experts may primarily stabilize conceptual abstractions, others may integrate contextual representations, and still others may perform transformations related to action-oriented outputs.

Under this interpretation, the expert layer of a modular neural architecture could be viewed as a stabilization landscape within the pathway space defined by the OPT framework. Mapping this landscape may provide insight into how reasoning functions are distributed across the model.

Future studies could explore this possibility by examining how expert activations correlate with different reasoning tasks, linguistic structures, or representational patterns.

This set of hypotheses highlights how the OPT framework may guide the interpretation of routing behaviour in modular neural architectures. Rather than treating expert selection as an opaque statistical process, the pathway perspective suggests that routing dynamics may reflect structured signal propagation within the model.

Although the present work does not attempt to empirically validate these hypotheses, they outline a potential research agenda for investigating the structural organization of routing mechanisms in large language models.

6 Implications for Interpretability and Architecture Design

Interpreting modular neural architectures through the OPT pathway framework suggests several implications for understanding and designing large language models. By treating routing behaviour as structured signal propagation rather than purely statistical gating, the framework provides a conceptual basis for analysing how reasoning trajectories emerge within large-scale neural systems.

The following sections outline several areas in which the pathway perspective may contribute to ongoing research on model interpretability and modular architecture design.

6.1 Pathway-Based Interpretability

One of the central challenges in contemporary language models is the interpretability of internal computation. Although modern models achieve impressive performance across many tasks, the internal mechanisms that produce their outputs often remain difficult to analyse.

Within the OPT framework, interpretability can be approached by examining the pathways through which signals propagate inside the model. Instead of analysing individual neurons or attention heads in isolation, the pathway perspective focuses on how signals move between generative processes and stabilization mechanisms.

In a modular architecture, routing patterns provide a natural observable proxy for these pathways. If routing distributions reflect stable signal propagation tendencies, then analysing expert activation patterns may reveal how the model organizes different forms of reasoning.

For example, certain experts may consistently process signals associated with conceptual abstraction, while others may stabilize contextual representations or task-oriented outputs. Observing such regularities could provide insight into how reasoning processes are distributed across the model.

This pathway-based view does not replace existing interpretability methods, but it suggests an alternative level of analysis that focuses on signal routing rather than on individual parameters.

6.2 Structured Routing for Modular Systems

The OPT framework also suggests possible directions for improving routing mechanisms in modular architectures. Current MoE systems typically rely on gating networks that learn routing policies through gradient optimization alone. While effective, such routing mechanisms often remain difficult to interpret and may produce unstable expert utilization patterns.

Introducing structural constraints inspired by pathway models could potentially improve the stability of modular systems. For instance, routing mechanisms might be designed to preserve consistent signal trajectories across reasoning steps rather than making independent expert selections at each layer.

In such a design, routing decisions would reflect not only local input features but also the broader trajectory of signal propagation within the model. This could allow modular systems to develop more coherent reasoning paths across multiple computational stages.

Although the present work does not propose a specific algorithm for implementing pathway-constrained routing, the OPT framework provides a conceptual basis for exploring such architectures.

6.3 Hybrid Cognitive – Computational Models

Finally, the pathway perspective highlights a possible convergence between cognitive modelling and machine learning architectures. Many theories of cognition emphasize that reasoning emerges from the interaction between generative processes and mechanisms that stabilize interpretations or actions.

The OPT framework formalizes this idea by describing cognition as the propagation of signals between generative sources and stabilization sinks. When applied to modular neural systems, this formulation suggests that large language models may already approximate similar structures through learned routing dynamics.

Viewing neural architectures through this lens may therefore provide a shared vocabulary for discussing reasoning processes across both biological and artificial systems. Rather than treating cognitive theory and machine learning as separate domains, pathway-based models may offer a structural language that connects them.

Such connections remain speculative at the present stage. However, the possibility of linking cognitive pathway models with modular neural architectures suggests a promising direction for future interdisciplinary research.

7 Conclusion

This study introduced the OPT framework as a structural model designed to explain why cognition unfolds through particular patterns rather than merely describing observable behavioural differences. Unlike traditional typological models that classify individuals into static categories, the OPT framework focuses on the mechanisms through which signals propagate between generative processes and stabilization mechanisms. By modelling cognition as a dynamic routing system, the framework shifts the analytical focus from classification to structural causality.

Within this perspective, cognitive activity is understood as the propagation of signals through pathways connecting generative sources and stabilization sinks. Generative processes initiate signals through abstraction or representation retrieval, while stabilization mechanisms integrate these signals into coherent interpretations, decisions, or actions. The interaction between these components produces the reasoning trajectories that shape cognitive behaviour.

Building on this conceptual model, the present work proposed that similar routing dynamics may also appear in modular neural architectures. In particular, routing mechanisms in Mixture-of-Experts language models can be interpreted as approximations of pathway selection processes within a distributed computational system. Expert modules may function as stabilization mechanisms that resolve signals generated by earlier layers of the model, while routing networks determine how signals propagate across these modules.

Although this interpretation does not imply that current neural architectures explicitly implement the OPT model, it suggests that pathway-based descriptions may provide a useful conceptual framework for analysing routing behaviour in large-scale neural systems. The pathway perspective offers a structural vocabulary for examining how reasoning processes emerge from the interaction between signal generation, routing, and stabilization.

Beyond theoretical interpretation, the framework also suggests potential directions for future research. Investigating expert specialization, routing stability, and modular topology may reveal whether pathway-like structures emerge within trained neural models. At the same time, pathway-based perspectives may inform the design of more interpretable routing mechanisms in modular architectures.

As the first formal presentation of the OPT framework, the present study establishes its conceptual foundations and outlines possible connections with contemporary machine learning systems. Further work will be required to evaluate its empirical implications and explore its applications across cognitive science, artificial intelligence, and interdisciplinary studies of reasoning processes. If these directions prove productive, pathway-based models may offer a new perspective on how complex reasoning systems organize and stabilize the flow of information.

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